



UNIVERSITY OF LEEDS

Formal Language & Finite Automata

Context-free grammars & Pushdown automata

Dr Noleen Koehler

University of Leeds

Context-free languages

The main aim of this week is to demonstrate that CFG's are equivalent to Pushdown automata.

To do this, we will

1. Show that for every language generated by a CFG there exists a PDA that recognises it. Done!
2. Show that for every language recognised by a PDA there exists a CFG that generates it. To be continued today!

**Show that for every language
recognised by a PDA there exists
a CFG that generates it.**

Pushdown automata $\implies$ Context-free grammar

Given a pushdown automata we construct a CFG.

Firstly we require the pushdown automata to have three properties,

1. It has a single accept state q_{accept} .
2. It empties it's stack before accepting.
3. Each transition either pushes a symbol onto the stack, or pops a symbol off the stack - but not both.

Pushdown automata $\implies$ Context-free grammar

Let $P = (Q, \Sigma, \Gamma, \delta, q_0, \{q_{\text{accept}}\})$ that meet the conditions specified previously, we construct a grammar G such that $L(G) = L(P)$. The grammar G is constructed as follows, where $G = (V, \Sigma, R, S)$.

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The variables of G are $\{A_{p,q} \mid p, q \in Q\}$. The start symbol of G is $A_{q_0, q_{\text{accept}}}$. The production rules are defined as follows:

1. For each $p, q, r, s \in Q$, $u \in \Gamma$ and $a, b \in \Sigma_\epsilon$, if $(r, u) \in \delta(p, a, \epsilon)$ and $(q, \epsilon) \in \delta(s, b, u)$ then $A_{pq} \rightarrow aA_{rs}b$ is a member of R .
2. For each $p, q, r \in Q$ we have $A_{pq} \rightarrow A_{pr}A_{rq}$ is a member of R
3. For each $p \in Q$ we have $A_{pp} \rightarrow \epsilon$ is a member of R

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We must prove that $A_{pq} \xRightarrow{*} x$ if and only if x can take P from state p with an empty stack to state q with an empty stack. Each direction of the proof is considered separately.